

(R&D) programmes. The government has opened testing facilities and military ranges for companies developing systems for defence applications. The government's R&D policy has made a huge and positive impact on India's defence hardware manufacturing sector.

Weapon testing includes testing of tensile strength of materials, reliability of equipment and other performance parameters. Use of government-owned testing facilities will result in lower cost in addition to faster development and production of defence equipment in India. This will reduce the dependency on foreign suppliers for testing and certification, saving both time and money. Some of the laboratories funded by the government are Defence Research and Development Organisation (DRDO), Bharat Electronics Ltd (BEL) and Hindustan Aeronautics Ltd (HAL).

Role of T&M in defence

Military devices are tested to work in extremely high temperatures, highly humid conditions, heavy rains and other special conditions. Test equipment and measurement solutions have assisted the rapid technological growth of the military. The T&M segment has experienced a huge demand to set up a secure defence communication network. Such equipment is mandated to meet MIL standards. These must be portable and battery-operated so that these can be taken to the field.

T&M vendors for defence applications need to fulfil application, performance, quality and support. Leading companies have been focused on supporting the total life cycle management of test equipment to improve their longevity. Evolving test needs have increased the importance of parameters other than accuracy. These parameters are scalability, flexibility, upgradability, interoperability and total cost of ownership, and require smarter test systems built on modular platforms that can scale to meet current and future needs.

The most important requirements for T&M in the defence industry are adaptability to evolving tech-



An Arjun MBT being test driven on the bump track at Central Vehicles Research and Development Establishment (CVRDE) at Avadi, Chennai (Credit: commons.wikimedia.org)

nologies, ability to address greater bandwidths and providing software-defined instrumentation. With growing complexity of end-user systems and testing needs, upgradable and flexible platforms are the focus of leading companies for enhancing the longevity of test equipment and reducing the cost of testing.

Devices in the market

The most important requirements in terms of test equipment are full capture of signals of greater bandwidth, wideband capabilities, architecture alignment with multifunction antenna systems, accurate information and greater ease of use. Applications such as electronic warfare, radar prototyping, spectrum monitoring and signal intelligence use modular test equipment, as these support radio frequency (RF) requirements at a low cost.

Some areas that require specialised T&M for military applications are radar test systems, radar cross section (RCS) measurement systems, GPS simulators, signal conditioning systems, control and simulation systems, frequency response analys-

ers, arbitrary waveform generators, spectrum analysers, signal analysers, audio analysers, RF signal analysers, RF power meters, oscilloscopes, logic analysers, multimeters, signal generators, network analysers, power meters, electronic counters, frequency counters, universal counters, and testing for electromagnetic compatibility (EMC), environmental, audio/video, military communication, radio, satellite and signals intelligence.

Moving large amounts of data quickly is particularly important for military applications. Antenna systems, Bluetooth connectivity and the Internet of Things (IoT) benefit from software-enabled instruments.

High-frequency T&M requirements

Continuous adoption of ultraband radars and giga-hertz bandwidth has become a key requirement for spectrum and signal analysers. The trend towards millimetre-wave frequencies and higher modulation bandwidths impacts the research for special-purpose radars and the high-frequency T&M market.

There is also a shift towards